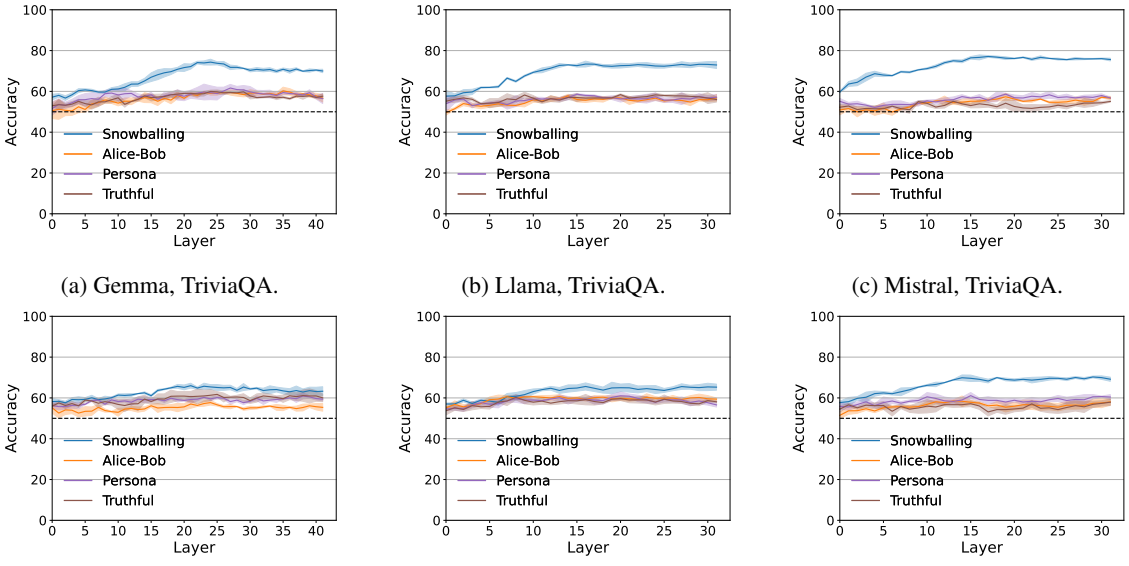


(a) Gemma. (b) Llama. (c) Mistral.

Figure 7: 3-way classification results on Natural Questions into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples. All the detection results are well above the random baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.



(a) Gemma, TriviaQA. (b) Llama, TriviaQA. (c) Mistral, TriviaQA.

(d) Gemma, Natural Questions. (e) Llama, Natural Questions. (f) Mistral, Natural Questions.

Figure 8: Classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+). We can see that all the detection lines are well above the random baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	14987	842	7356
Natural-Questions-Llama3-WACK	6203	835	14739
TriviaQA-Gemma-WACK	15458	639	6991
Natural-Questions-Gemma-WACK	7253	651	13762
TriviaQA-Mistral-WACK	14728	765	7650
Natural-Questions-Msitral-WACK	6421	687	14689

Table 11: Dataset label statistics on the Truthful setting.

Model	Data Set	HK^+	HK^-
Gemma	TriviaQA	**15.4 / 20.0 / 19.6 / 7.8**	2.2 / 1.6 / 1.8 / 2.2
	Natural Questions	**17.2 / 23.2 / 19.0 / 11.6**	2.2 / 1.0 / 1.6 / 1.8
Llama	TriviaQA	**18.6 / 26.2 / 16.4 / 11.6**	1.4 / 1.6 / 0.6 / 2.2
	Natural Questions	**16.8 / 25.4 / 20.2 / 18.4**	1.0 / 1.4 / 0.6 / 1.0
Mistral	TriviaQA	**24.6 / 17.4 / 18.6 / 18.2**	1.8 / 3.2 / 1.0 / 3.4
	Natural Questions	**23.6 / 20.0 / 15.4 / 16.4**	1.4 / 2.2 / 0.8 / 1.4

Table 12: Comparison of HK^+ and HK^- mitigation results using alternative mitigation prompt on Truthful / Persona / Alice-Bob / Snowballing settings. We can see that HK^+ mitigation is significantly higher than HK^-.

specific settings used (e.g., Alice-Bob, Truthful, etc.).

Figure 9 presents additional results for configurations beyond those discussed in the main paper. Specifically, Figures 9a and 9b illustrate the similarity of settings under the Gemma and Metta models, while Figures 9c and 9d highlight the hallucination similarity across models using Alice-Bob and Truthful settings. These findings mirror the patterns observed in the main paper and further underscore the differences between models and settings. This emphasizes the necessity of developing model- and setting-specific datasets.

G HK^+ Detection Additional Results

To complete the results from Section 6.1 on TriviaQA, we show in Figure 10 similar detection results that compare generic and model's specific dataset on Natural Questions.

H Preemptive Hallucination Detection Additional Results

In Section 6.2, we demonstrate that hallucinations can be detected before they are generated, whereas post-hallucination settings are less effective for this purpose. While the main paper presents results using the Mistral model, we extend this evaluation to Llama (Figure 11) and Gemma (Figure 12). The results for these models exhibit similar patterns to those observed with Mistral, reinforcing the generality of this phenomenon.

I Detection results on the MLP and Attention Components

The results in the main paper are only shown using hidden states from the residual component of the LLMs, that is, the representations after each transformer block. To complete the picture, we provide detection results also for the MLP and attention components using the representations that are output by the component.

The results are shown in Figures 13 and 14 for the classification into the two hallucination types and factually correct examples, for all the combinations of datasets and models.

Next in Figures 15 and 16 we see the results of the generalization of the Snowballing setting to the Alice-Bob setting using the MLP and Attention components.

Lastly, Figures 17 and 18 show detection of HK^+ results using classifiers trained on specific and generic datasets.

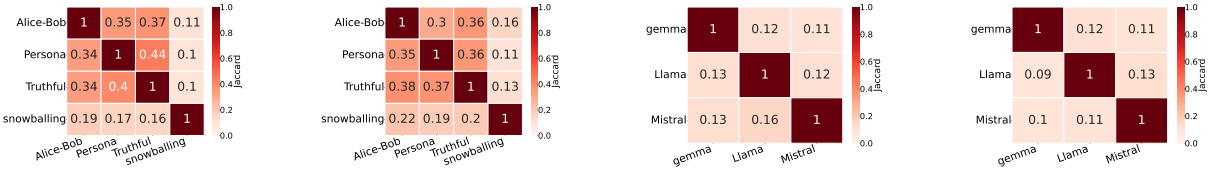
In all these figures, the results with the MLP and attention components yield similar trends to the ones in the main paper using the residual component, albeit with moderately lower accuracy. This implies that the detection results are not limited to a specific component and are a broader phenomenon across components.

J Generalization between Settings

In Section 5.2, we demonstrated that the Snowballing setting is generalizable to the Alice-Bob setting. To further validate this generalization, we present Figures 19 and 20, which show that Snowballing is also generalizable to the Persona and Truthful settings.

K Different snowballing Configuration have Similar Results

In the main paper, we used a single configuration with 3 bad shots, randomly selected. To demonstrate that our results are not dependent on this specific configuration, we conducted additional experiments with a different Snowballing setting. Specifically, we increased the number of shots to five and used a different random seed in the dataset creation process to sample differ-



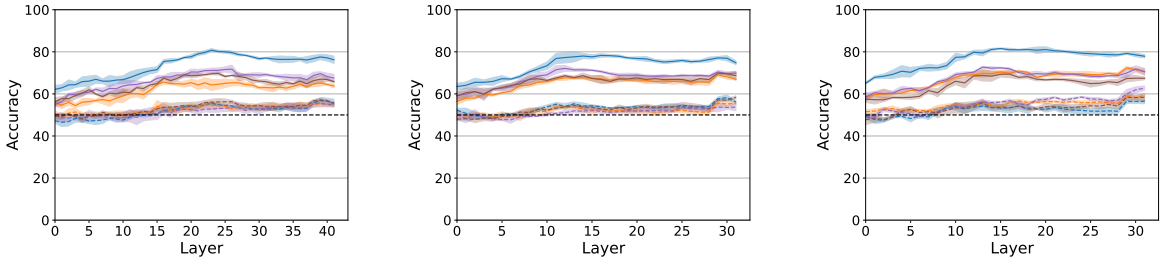
(a) Gemma HK^+ similarity between settings.

(b) Llama HK^+ similarity between settings.

(c) HK^+ similarity under Alice-Bob.

(d) HK^+ similarity under Truthful setting.

Figure 9: High-Knowledge and HK^+ differences on TriviaQA (above the diagonal) and Natural Questions (below the diagonal) between the models and settings. We can see that different models have different knowledge and different HK^+ examples in shared knowledge.



(a) Gemma.

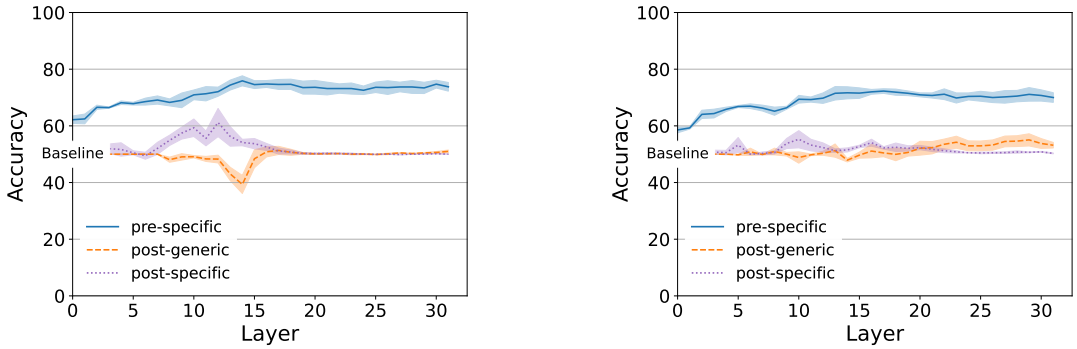
(b) Llama.

(c) Mistral.

Figure 10: Distinguishing factually correct from HK^+ using classifiers trained on generic vs. model-specific datasets on Natural Questions. We can see that specific dataset accuracy is significantly higher than generic dataset.

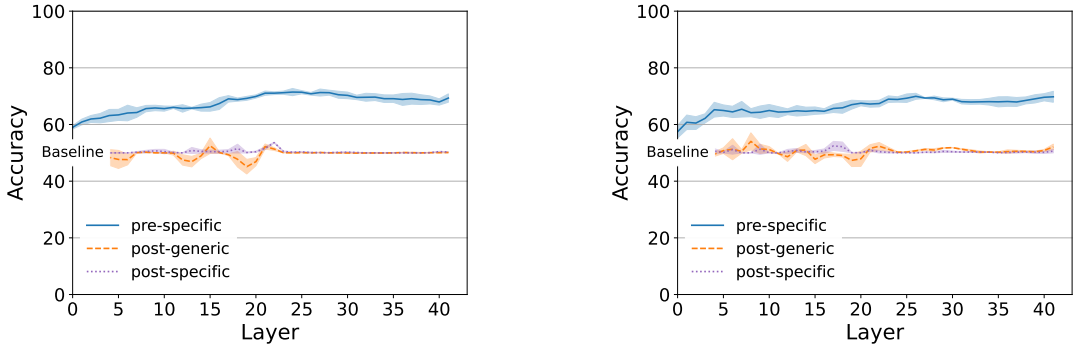
ent Snowballing. Our main objective was to show that the classifier trained under this new configuration performs similarly to the original 3-Snowballing-shot setting. Demonstrating this would suggest that variations in Snowballing configurations do not impact classifier performance.

In Figure 21 we present the performance of a classifier trained in the 5-Snowballing-shot configuration to distinguish between factually correct answers, HK^-, and HK^+. These results are comparable to those reported in the main paper in Section 4.2 for the Snowballing configurations. This consistency indicates that the 5-Snowballing-shots configuration does not affect the classifier's ability to differentiate between the various types of hallucinations.



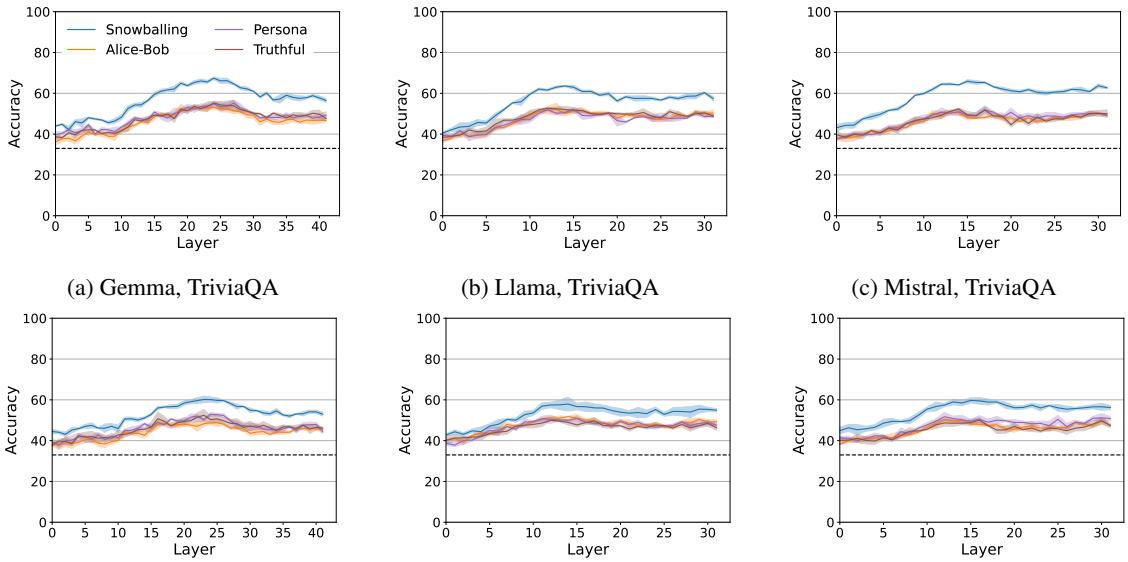
(a) TriviaQA (b) Natural Questions

Figure 11: Comparing HK^+ detection before generation using classifiers trained on model-specific pre-hallucination, generic post-hallucination, and model-specific post-hallucination examples on Llama. We can see that preemptive hallucination is effective and that generic dataset can not detect it.



(a) TriviaQA (b) Natural Questions

Figure 12: Comparing HK^+ detection before generation using classifiers trained on model-specific pre-hallucination, generic post-hallucination, and model-specific post-hallucination examples on Gemma. We can see that preemptive hallucination is effective and that generic dataset can not detect it.



(a) Gemma, TriviaQA (b) Llama, TriviaQA (c) Mistral, TriviaQA

(d) Gemma, Natural Questions (e) Llama, Natural Questions (f) Mistral, Natural Questions

Figure 13: 3-way classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples on the MLP component. We can see that all the detection lines are well above the baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.